

Algebra 1
Unit 9
Lesson 11 Practice

Name Answer Key
Date _____
Period _____

A barber in Gainesville, Florida can track his expected revenue, $R(x)$, in dollars, each week based on what he charges for a basic haircut. The table shows the revenue model for the barber, based on x , the cost for a basic haircut.

↓ vertex

Cost, x	2	4	6	8	10	12	14
Revenue, \$ $R(x)$	-100	2700	4700	5900	6300	5900	4700

1. Write a quadratic equation, in vertex form, that models the revenue, $R(x)$, in terms of the basic haircut's cost, x .

$$\begin{aligned}
 y &= a(x - 10)^2 + 6300 \\
 -100 &= a(2 - 10)^2 + 6300 \\
 -100 &= 64a + 6300 \\
 -6400 &= 64a \\
 -100 &= a
 \end{aligned}
 \qquad
 y = -100(x - 10)^2 + 6,300$$

2. Write the quadratic equation in standard form.

$$\begin{aligned}
 y &= -100(x - 10)(x - 10) + 6300 \\
 y &= -100(x^2 - 20x + 100) + 6300 \\
 y &= -100x^2 + 2000x - 10000 + 6300 \\
 y &= -100x^2 + 2,000x - 3,700
 \end{aligned}$$

3. Use the quadratic equation to determine how much the barber should charge for a basic haircut to estimate that the revenue coming that week is \$6075.

$$\begin{aligned}
 6075 &= -100(x - 10)^2 + 6300 & x - 10 &= \pm \frac{15}{10} \\
 -225 &= -100(x - 10)^2 & x &= \$11.50 \\
 \frac{225}{100} &= (x - 10)^2 & x &= \$8.50
 \end{aligned}$$

The table models the height, $h(x)$, in feet, after x seconds, of a toy projectile that is launched from 22 feet above the ground.

vertex ↓

Seconds, x	1	2	3	4	5
Height, $h(x)$	85	130	157	166	157

4. Write a quadratic equation, in vertex form, that models the height of the toy in terms of seconds, x .

$$\begin{aligned}
 y &= a(x - 4)^2 + 166 \\
 130 &= a(2 - 4)^2 + 166 \\
 -36 &= 4a \\
 -9 &= a
 \end{aligned}
 \qquad
 y = -9(x - 4)^2 + 166$$

5. Rewrite the quadratic equation modeling the toy's height in standard form.

$$y = -9(x^2 - 8x + 16) + 166$$

$$y = -9x^2 + 72x - 144 + 166$$

$$y = -9x^2 + 72x + 22$$

6. Use the quadratic equation to determine at what time the toy will land back on the ground.

$$0 = -9(x - 4)^2 + 166$$

$$\pm \sqrt{\frac{166}{9}} = x - 4$$

$$-166 = -9(x - 4)^2$$

$$x = 4 \pm \sqrt{\frac{166}{9}}$$

$$\frac{-166}{-9} = (x - 4)^2$$

$$x \approx -0.295, \underline{x \approx 8.29 \text{ seconds}}$$

The function, $M(x)$, models the number of miles, in millions, that U.S. passengers traveled on a train for the period from 1970 to 2000, where x is the number of years since 1970.

7. Write a quadratic equation, in vertex form, that models the miles traveled by U.S. train passengers, $M(x)$, in terms of years since 1970, x .

$$y = a(x - 16)^2 + 4,899$$

$$4919 = a(14 - 16)^2 + 4899$$

$$20 = 4a$$

$$5 = a$$

$$y = 5(x - 16)^2 + 4,899$$

x	$M(x)$
14	4919
15	4904
16	4899
17	4904
18	4919
19	4944

8. Rewrite the quadratic equation in standard form.

$$y = 5(x^2 - 32x + 256) + 4899$$

$$y = 5x^2 - 160x + 1280 + 4899$$

$$y = 5x^2 - 160x + 6,179$$

9. Use the quadratic equation to determine when $y = 6179$ miles.

$$6179 = 5(x - 16)^2 + 4,899$$

$$1280 = 5(x - 16)^2$$

$$256 = (x - 16)^2$$

$$\pm 16 = x - 16$$

$$-16 = x - 16$$

$$x = 0$$

$$16 = x - 16$$

$$x = 32$$

10. Interpret the solutions and determine if the solutions are viable in context.

In 1970, U.S. passengers travelled 6,179 miles. $x = 32$ is not viable because this is 2002, which is outside the domain of the model.